Unit outline_

COMP5348: Enterprise Scale Software Architecture

Semester 2, 2026 [Normal evening] - Camperdown/Darlington, Sydney

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Overview –

This unit covers topics on software architecture for large-scale enterprises. Computer systems for large-scale enterprises handle critical business processes, interact with computer systems of other organisations, and have to be highly reliable, available and scalable. This class of systems are built up from several application components, incorporating existing legacy code and data stores as well as linking these through middleware technologies, such as distributed transaction processing, remote objects, message-queuing, publish-subscribe, and clustering. The choice of middleware can decide whether the system achieves essential non- functional requirements such as performance and availability. The objective of this unit of study is to educate students for their later professional career and it covers Software Architecture topics of the ACM/IEEE Software Engineering curriculum. Objective: The objective of this unit of study is to educate students for their later professional career and it covers topics of the ACM/IEEE Software Engineering curriculum.

Unit details and rules

Academic unit	Computer Science
Credit points	6
Prerequisites ?	None
Corequisites ?	None
Prohibitions ?	COMP4348
Assumed knowledge ?	Experience with software development as covered in SOFT2412 or COMP9103 and also COMP2123 or COMP2823 or INFO1105 or INFO1905 (or equivalent UoS from different institutions)
Available to study abroad and exchange students	Yes

Teaching staff

Coordinator	Sichao Li, sichao.li@sydney.edu.au

Assessment –

ⓘ The census date for this unit availability is **31 August 2026**

Details Criteria

Type	Description	Weight	Due	Length	Use of AI
Written exam 🚧	**Final exam** Written questions covering all lectures, labs and assessments.	60%	Formal exam period	2 hours	AI prohibited
Outcomes assessed: LO1 LO3 LO4 LO5 LO6					
Practical skill	**Assignment 1** Paper-based exercises based on material of weeks 1-6.	10%	Week 07	n/a	AI allowed
Outcomes assessed: LO3 LO4 LO5 LO6					
Practical skill	**Assignment 2** Paper-based tasks on W7-W10 materials, & reflecting on labs experiences.	10%	Week 11	n/a	AI allowed
Outcomes assessed: LO1 LO4 LO5 LO6					
Creative work 👥	**Group project** Practical work integrating subsystems using communication technologies.	10%	Week 12	n/a	AI allowed
Outcomes assessed: LO1 LO2 LO4 LO5					
Presentation 👥	**Group project presentation** It is a group presentation based on the Group Project.	10%	Week 13	15 mins during tutorials.	AI allowed
Outcomes assessed: LO1 LO2 LO3 LO4 LO5 LO6					

🚧 = hurdle task ? 👥 = group assignment ?

Assessment summary

The assessments are designed to evaluate students' ability to design, analyse, and implement enterprise software architectures. They assess essential knowledge and skills in state management, concurrency, transactions, distributed computing, middleware, performance evaluation, web development, and cloud computing.

Assignment 1 is a paper-based assessment covering material from Weeks 1–6.

Assignment 2 is a paper-based assessment covering material from Weeks 7–10.

Group project is a practical subsystem integration project involving software implementation, system integration, teamwork, and presentation.

The **final exam** consists of written questions and covers all lectures, labs, and assessment-related material.

Use of generative artificial intelligence (AI)

You can use generative AI tools for open assessments. Restrictions on AI use apply to secure, supervised assessments used to confirm if students have met specific learning outcomes.

Refer to the assessment table above to see if AI is allowed, for assessments in this unit and check Canvas for full instructions on assessment tasks and AI use.

If you use AI, you must always acknowledge it. Misusing AI may lead to a breach of the Academic Integrity Policy.

Visit the Current Students website for more information on AI in assessments, including details on how to acknowledge its use.

Late submission

In accordance with University policy, these penalties apply when written work is submitted after 11:59pm on the due date:

- Deduction of **5%** of the maximum mark for each calendar day after the due date.
- After ten calendar days late, a mark of zero will be awarded.

This unit has an exception to the standard University policy or supplementary information has been provided by the unit coordinator. This information is displayed below:

Late submissions will not be accepted unless the student has been granted Special Consideration in accordance with University policy.

Academic integrity

The University expects students to act ethically and honestly and will treat all allegations of academic integrity breaches seriously.

Our website provides information on academic integrity and the resources available to all students. This includes advice on how to avoid common breaches of academic integrity. Ensure that you have completed the Academic Honesty Education Module (AHEM) which is mandatory for all commencing coursework students

Penalties for serious breaches can significantly impact your studies and your career after graduation. It is important that you speak with your unit coordinator if you need help with completing assessments.

Visit the Current Students website for more information on AI in assessments, including details on how to acknowledge its use.

Learning support –

Simple extensions

If you encounter a problem submitting your work on time, you may be able to apply for an extension of five calendar days through a simple extension. The application process will be different depending on the type of assessment and extensions cannot be granted for some assessment types like exams.

Special consideration

If exceptional circumstances mean you can't complete an assessment, you need consideration for a longer period of time, or if you have essential commitments which impact your performance in an assessment, you may be eligible for special consideration or special arrangements.

Special consideration applications will not be affected by a simple extension application.

Using AI responsibly

Co-created with students, AI in Education includes lots of helpful examples of how students use generative AI tools to support their learning. It explains how generative AI works, the different tools available and how to use them responsibly and productively.

Support for students

The Support for Students Policy reflects the University's commitment to supporting students in their academic journey and making the University safe for students. It is important that you read and understand this policy so that you are familiar with the range of support services available to you and understand how to engage with them.

The University uses email as its primary source of communication with students who need support under the Support for Students Policy. Make sure you check your University email regularly and respond to any communications received from the University.

Learning resources and detailed information about weekly assessment and learning activities can be accessed via Canvas. It is essential that you visit your unit of study Canvas site to ensure you are up to date with all of your tasks.

If you are having difficulties completing your studies, or are feeling unsure about your progress, we are here to help. You can access the support services offered by the University at any time:

Support and Services (including health and wellbeing services, financial support and learning support)
Course planning and administration
Meet with an Academic Adviser

Weekly schedule –

WK	Topic	Learning activity	Learning outcomes
Week 01	Introduction	Lecture (1 hr)	LO1 LO2 LO3 LO4 LO5 LO6
	Introduction	Tutorial (2 hr)	LO1 LO2 LO3 LO4 LO5 LO6
Week 02	State and concurrency	Lecture (1 hr)	LO1 LO2 LO3 LO4 LO5 LO6
	State and concurrency	Tutorial (2 hr)	LO1 LO2 LO3 LO4 LO5 LO6
Week 03	Distributed computing	Lecture (1 hr)	LO1 LO2 LO3 LO4 LO5 LO6
	Distributed computing	Tutorial (2 hr)	LO1 LO2 LO3 LO4 LO5 LO6
Week 04	Distributed transactions	Lecture (1 hr)	LO1 LO2 LO3 LO4 LO5 LO6
	Distributed transactions	Tutorial (2 hr)	LO1 LO2 LO3 LO4 LO5 LO6
Week 05	Performance	Lecture (1 hr)	LO1 LO2 LO3 LO4 LO5 LO6
	Performance	Tutorial (2 hr)	LO1 LO2 LO3 LO4 LO5 LO6
Week 06	Message oriented middleware and asynchronous technologies	Lecture (1 hr)	LO1 LO2 LO3 LO4 LO5 LO6
	Message oriented middleware and asynchronous technologies	Tutorial (2 hr)	LO1 LO2 LO3 LO4 LO5 LO6
Week 07	Service-oriented architecture and RESTful services	Lecture (1 hr)	LO1 LO2 LO3 LO4 LO5 LO6
	Service-oriented architecture and RESTful services	Tutorial (2 hr)	LO1 LO2 LO3 LO4 LO5 LO6
Week 08	Predicting performance and queues	Lecture (1 hr)	LO1 LO2 LO3 LO4 LO5 LO6
	Predicting performance and queues	Tutorial (2 hr)	LO1 LO2 LO3 LO4 LO5 LO6
Week 09	Scalability, availability and fault-tolerance	Lecture (1 hr)	LO1 LO2 LO3 LO4 LO5 LO6
	Scalability, availability and fault-tolerance	Tutorial (2 hr)	LO1 LO2 LO3 LO4 LO5 LO6
Week 10	Professional pathway towards software architect	Lecture (1 hr)	LO1 LO2 LO3 LO4 LO5 LO6
	Professional pathway towards software architect	Tutorial (2 hr)	LO1 LO2 LO3 LO4 LO5 LO6
Week 11	Software architecture in practice	Lecture (1 hr)	LO1 LO2 LO3 LO4 LO5 LO6
	Software architecture in practice	Tutorial (2 hr)	LO1 LO2 LO3 LO4 LO5 LO6
Week 12	Cloud computing	Lecture (1 hr)	LO1 LO2 LO3 LO4 LO5 LO6
	Cloud computing	Tutorial (2 hr)	LO1 LO2 LO3 LO4 LO5 LO6
Week 13	Review and exam preparation	Lecture (1 hr)	LO1 LO2 LO3 LO4 LO5 LO6
	Review and exam preparation	Tutorial (2 hr)	LO1 LO2 LO3 LO4 LO5 LO6

Attendance and class requirements

- **Lectures**: face-to-face and online lecture recording on Canvas.
- **Laboratory:** Laboratory sessions are for students to practice problem-solving, application of the concepts, and use of technologies.
- **Independent study**: Independent study is required for reading from sources, doing assignments, etc

Study commitment

Typically, there is a minimum expectation of 1.5-2 hours of student effort per week per credit point for units of study offered over a full semester. For a 6 credit point unit, this equates to roughly 120-150 hours of student effort in total.

Required readings

Note: References are provided for guidance purposes only. Students are advised to consult these books in the university library. Purchase is not required.

- Essential Software Architecture
- Microsoft Application Architecture Guide (2nd ed)
- Performance By Design
- REST in Practice (Hypermedia and Systems Architecture)
- Software Architecture in Practice

Learning outcomes –

Learning outcomes are what students know, understand and are able to do on completion of a unit of study. They are aligned with the University's graduate qualities and are assessed as part of the curriculum.

Outcomes Graduate qualities

At the completion of this unit, you should be able to:

LO1. Produce written evaluation of different architectures and/or of different technologies.
LO2. Work in small teams of people with diverse skills and backgrounds, to complete demanding tasks which are loosely defined and require rapid learning of new concepts.
LO3. Understand the role of a software architect; respect what he/she is doing, and why; know when to involve him/her.
LO4. Recognise the relationship between different software architecture choices, and their impact on various non-functional attributes of the software.
LO5. Demonstrate broad knowledge of common architectural approaches for enterprise software, as well as detailed skills working with some technologies available to implement those approaches.
LO6. Carry out performance analysis, explain performance measurement principles, and identify suitable approaches for state management in face of concurrency.

Responding to student feedback –

This section outlines changes made to this unit following staff and student reviews.

I have made adjustments to better align the unit with the University's academic calendar and improve the student experience. The assessments have also been updated to reflect recent developments in generative AI, support academic integrity, and mitigate the risk of over-reliance on AI-generated work.

Disclaimer

Important: the University of Sydney regularly reviews units of study and reserves the right to change the units of study available annually. To stay up to date on available study options, including unit of study details and availability, refer to the relevant handbook.

The information included on this page is correct at 10 Jul 2026.

To help you understand common terms that we use at the University, we offer an online glossary.

We recognise and pay respect to the Elders and communities - past, present, and emerging - of the lands that the University of Sydney's campuses stand on. For thousands of years they have shared and exchanged knowledges across innumerable generations for the benefit of all.

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